

CONTACT

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RESEARCH INTERESTS

Robotic Manipulation, Vision-guided Robotic Manipulation, Constrained Motion Planning, Robot Learning, Task and Motion Planning.

EDUCATION

Stony Brook University, Stony Brook, NY, USA.

Ph.D., Mechanical Engineering - Robotics, minor in Computer Science (August 2019 - July 2026)

Thesis Title : Sensor-Based Task-Oriented Grasping [\[pdf\]](#)

Advisor : [Dr. Nilanjan Chakraborty](#)

Stony Brook University, Stony Brook, NY, USA.

M.S., Mechanical Engineering (August 2017 - August 2019)

Thesis Title : Grasping Force Optimization while considering Hand Capabilities and Object-Environment Contact

Advisor : [Dr. Nilanjan Chakraborty](#)

Savitribai Phule Pune University, Pune, Maharashtra, India.

B.E., Mechanical Engineering - *First Class with Distinction* (August 2012 - June 2016)

PROFESSIONAL EXPERIENCE

Stony Brook University, Stony Brook, NY, USA.

Postdoctoral Associate - [The Soft Flyers Group](#) (August 2026 - Present)

- Research on Multimodal Aerial and Underwater Vehicles.
- Research on Robotic Manipulation and Planning.

Stony Brook University, Stony Brook, NY, USA.

Research Assistant - [Interacting Robotic Systems Laboratory](#) (June 2020 - July 2026)

- **Grasping and Optimization:** Formulated a novel task-dependent grasp metric as a **Second-Order Cone Program (SOCP)** to evaluate grasps (antipodal and multifingered) based on their feasibility to impart the desired motion after grasping. Proposed formulation incorporates object-environment contact and dynamics **without approximating the friction cone**.
- **Grasp Synthesis:** Developed novel analytical and data-driven approaches for task-oriented grasp (regrasp) synthesis directly from **sensor data** while considering the motion to be imparted to the object.
- **Constrained Planning:** Developed novel motion planning, force planning, and trajectory optimization approaches incorporating environmental contact constraints. Research on utilizing the screw-geometric structure of motion for constrained manipulation, providing robots with self-evaluation capabilities, incremental acquisition of kinesthetic demonstrations, motion generalization and task planning.
- **Perception and Manipulation:** Engineered a perception and manipulation stack using **ROS** in **Python** and **C++**, integrating state-of-the-art deep learning-based perception alongside novel screw geometry-based methods for manipulation.
- **Foundation Models for Task and Motion Planning:** Exploring the integration of foundation models, such as **VLMs** and **LLMs**, with screw geometry-based approaches to enhance a robot's task planning capabilities, particularly for tasks involving end-effector motion constraints.
- **Dexterous Manipulation:** Conducting active research to develop dexterous manipulation capabilities for mobile manipulators and humanoids, by bridging geometric constraint modeling with data-driven approaches.

GE Aerospace Research, Niskayuna, NY, USA.

Intern - Robotics and Embodied AI PhD Fellow (June 2025 - August 2025)

- **NVIDIA Robotics Stack Validation:** Evaluated state-of-the-art NVIDIA algorithms for High-Mix Low-Volume industrial automation. Specifically validated **cuRobo** for collision-free motion planning and **FoundationPose** within Isaac Sim for 6D pose estimation.
- **Synthetic Data Generation:** Developed a procedural defective part generation pipeline to evaluate perception systems for defect detection. Engineered a Python package to synthetically inject realistic "bubble-like" defects onto a CAD model of a ceramic-composite platform in simulation, enabling the rigorous testing of perception and planning algorithms.

- **Autonomous Scanning:** Designed a perception-driven motion planning framework for scanning tubes and ducts. Integrated **SAM** (Segment Anything Model) for segmentation to extract point clouds and compute an Oriented Bounding Box (OBB) to assign a local object reference frame. Integrated **ScLERP-based motion planning** to track 6-DoF scanning poses generated using the bounding box.

Nokia Bell Labs, Murray Hill, NJ, USA.

Research Intern - Robotics and 3D Modelling (June 2024 - August 2024)

Hosted by: [Dr. Matthew Andrews](#) and [Dr. Jeongran Lee](#) (Modelling and Optimization Group, AIRL)

- Studied the problem of representing and understanding large indoor physical spaces using radiance field techniques (NeRFs and 3D Gaussian Splatting) for developing corresponding **digital twins**.
- Developed pipelines for generating **3D reconstructions of large indoor environments (warehouses)** using data collected from fisheye cameras (Qoocam, Insta360, Kodak Pixpro SP360), ceiling cameras (Axis), and RGB-D cameras (Intel RealSense D435i) mounted on a mobile robot.
- Diagnosed and resolved key issues in camera calibration and data acquisition from RGB-D cameras on a moving mobile robot, leading to a **30%** improvement in PSNR during training for radiance field generation.

JOURNAL PUBLICATIONS

[A. Patankar](#), D. Mahalingam, A. Fakhari, and N. Chakraborty. **Task-Dependent Grasp Metric for Grasp Evaluation and Synthesis**. Under Preparation

D. Vahid, M. Boroji, [A. Patankar](#), N. Chakraborty, I. Kao, A. Fakhari. **Manipulation Planning by Exploiting Environmental Contact using Screw Theory**. Under Preparation

T. Zaw, D. Mahalingam, N. Baiata, [A. Patankar](#), N. Chakraborty. **Design and Kinematic Analysis of a Novel Series-Parallel Hybrid Finger for Robotic Hands**. *ASME JMR*, 2025. [[pdf](#)] [[video](#)]

CONFERENCE PUBLICATIONS

D. Das, [A. Patankar](#), N. Chakraborty, C.R. Ramakrishnan, and I.V. Ramakrishnan. **Screw Geometry meets Bandits: Incremental Acquisition of Demonstrations to Generate Manipulation Plans**. *Accepted at IEEE ICRA*, Vienna, 2026. [[arxiv](#)] [[video](#)]

D. Das, [A. Patankar](#), N. Chakraborty, C.R. Ramakrishnan, and I.V. Ramakrishnan. **Transferring Kinesthetic Demonstrations across Diverse Objects for Manipulation Planning**. *IEEE/RSJ IROS*, Hangzhou, 2025. [[pdf](#)] [[video](#)]

[A. Patankar](#), D. Mahalingam, and N. Chakraborty. **Synthesizing Grasps and Re-grasps for Complex Manipulation Tasks**. *IEEE ICRA*, Atlanta, USA, 2025. [[pdf](#)] [[video](#)]

K. Phi, [A. Patankar](#), D. Mahalingam, N. Chakraborty and I.V. Ramakrishnan. **Point Cloud Decomposition for Task-Oriented Grasping**. *IEEE ICRA*, Atlanta, USA, 2025.

D. Mahalingam, [A. Patankar](#), R. Laha, S. Lakshminarayanan, S. Haddadin and N. Chakraborty. **A General Formulation for Path Constrained Time-Optimized Trajectory Planning with Environmental and Object Contacts**. *IEEE/RSJ IROS*, Abu Dhabi, 2024. [[pdf](#)] [[video](#)]

D. Mahalingam, [A. Patankar](#), K. Phi, N. Chakraborty, R. McGann and I.V. Ramakrishnan. **Containerized Vertical Farming Using Cobots**. *IEEE ICRA*, Yokohama, Japan, 2024. [[pdf](#)] [[video](#)]

[A. Patankar](#), K. Phi, D. Mahalingam, N. Chakraborty and I.V. Ramakrishnan. **Task-Oriented Grasping with Point Cloud Representation of Objects**. *IEEE/RSJ IROS*, Detroit, USA, 2023. [[pdf](#)] [[webpage](#)]

A. Fakhari, [A. Patankar](#), J. Xie and N. Chakraborty. **Computing a Task-Dependent Grasp Metric Using Second-Order Cone Programs**. *IEEE/RSJ IROS*, Prague, Czech Republic, 2021. [[pdf](#)] [[video](#)]

A. Fakhari, [A. Patankar](#) and N. Chakraborty. **Motion and Force Planning for Manipulating Heavy Objects by Pivoting**. *IEEE/RSJ IROS*, Prague, Czech Republic, 2021. [[pdf](#)] [[video](#)]

[A. Patankar](#), A. Fakhari and N. Chakraborty. **Hand-Object Contact Force Synthesis for Manipulating Objects by Exploiting Environment**. *IEEE/RSJ IROS*, Las Vegas, NV, USA, 2020. [[pdf](#)] [[video](#)]

WORKSHOP PUBLICATIONS

K. Phi, A. T. Ho, [A. Patankar](#), N. Chakraborty, and I.V. Ramakrishnan. **Active Perception for Autonomous Multi-View Geometrically Consistent Data Collection in Local Environments**. Workshop on Learning and Representations for Active Perception in Manipulation, *IEEE ICRA*, Vienna, Austria, 2026. [[pdf](#)]

D. Mahalingam, [A. Patankar](#), D. Das, N. Chakraborty, and I.V. Ramakrishnan. **Caregiver-Guided Robotic Manipulation to Promote Independence of People with Significant Locomotor Disability**. Workshop on Assistive Robotics for Citizens, *IEEE/RSJ IROS*, Detroit, USA, 2023. [[pdf](#)]

D. Das, A. Patankar, F. Honda, D. Mahalingam, N. Chakraborty, C.R. Ramakrishnan, and I.V. Ramakrishnan. **Knowledge-Enabled Motion Generation for Complex Manipulation Tasks**. Workshop on Geometric Representations, *IEEE ICRA*, London, 2023. [\[pdf\]](#) [\[video\]](#)

PREPRINTS AND POSTERS

A. Patankar, D. Mahalingam, and N. Chakraborty. **Task-Dependent Grasp Metric with Object and Manipulator Dynamics**. Late Breaking Results Poster, *IEEE ICRA*, Atlanta, USA, 2025. [\[poster\]](#)

TECHNICAL SKILLS

Languages : Python, C++, MATLAB, Prolog
Frameworks : PyTorch, ROS, git, TensorFlow
Application Software and CAD : Inkscape, Blender, AutoCAD, Solidworks
Simulation : PyBullet, Gazebo, Isaac Sim, CoppeliaSim, Isaac Lab
Optimization : cvx, cvxpy, YALMIP, Gurobi
Perception and Processing : OpenCV, Open3D, nerfstudio, PCL, trimesh

PROFESSIONAL ACTIVITIES

Reviewer

IEEE Transactions on Robotics : 2021
 IEEE Transactions on Mechatronics : 2025
 IEEE International Conference on Robotics and Automation : 2020 - Present
 IEEE/RSJ International Conference on Intelligent Robots and Systems : 2020, 2024, 2025

Session Chair

Grasping - IEEE/RSJ International Conference on Intelligent Robots and Systems : 2021

MENTORING EXPERIENCE

Actively participated in mentoring high-school, undergraduate and graduate students

Simmons Summer Research Program - Stony Brook University : 2021
 Garcia Summer Research Program - Stony Brook University : 2021 - 2023
 CSE 523-24 Advanced Project in Computer Science : 2021 - Present

LEADERSHIP EXPERIENCE

Graduate Student Organisation - Secretary-elect : July 2020 - May 2021
Graduate Student Organisation - Senator : Sep 2019 - May 2020
Graduate Housing Residents Association - Treasurer-elect : Sept 2019 - Sept 2021
Indian Graduate Student Association - President-elect : Nov 2017 - Dec 2018

HONORS & AWARDS

Winner - Graduate Research Symposium, Stony Brook University, March 2023 & 2025
 Topic - ‘Task-Oriented Grasping with Point Cloud Representation of Objects’

Runner Up - SAE Baja India Technology Innovation Award 2016

Ranked 34th in the Maharashtra Talent Search Examination 2008 conducted by Modern Education Society’s Center for Talent Search and Excellence.

REFERENCES

Nilanjan Chakraborty

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